

OAK RIDGE, TN, USA

WMO: 724270

Lat: 36.023N

Lon: 84.234W

Elev: 900

StdP: 14.22

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53868

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.9	21.0	2.5	6.5	23.0	6.8	8.1	26.3	15.1	52.3	12.6	46.9	2.8	60	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	92.9	73.8	90.9	73.6	89.0	73.0	77.4	86.9	76.3	85.5	75.3	84.1	4.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.8	134.7	81.1	73.5	129.1	79.9	72.7	125.4	79.2	41.3	86.8	40.2	85.6	39.2	84.6	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.6	9.2	10.8	95.8	4.7	3.3	7.4	98.1	4.7	100.0	2.0	101.9	-1.4	104.2		
(5)			9.2	78.7	4.4	1.2	6.1	79.6	3.5	80.3	1.1	81.0	-2.1	81.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.9	38.5	42.8	50.9	60.4	68.5	75.7	78.4	78.0	72.1	60.6	49.6	42.2	
	DBStd	16.00	10.28	9.28	9.16	7.89	6.70	4.29	3.71	3.76	5.94	8.08	8.34	8.88	
	HDD50	1123	380	236	102	12	0	0	0	0	0	12	106	275	
	HDD65	3479	822	623	443	177	45	1	0	0	14	181	465	708	
	CDD50	4730	23	34	130	324	573	770	880	868	662	340	94	32	
	CDD65	1612	0	1	6	39	153	321	415	403	226	45	3	0	
	CDH74	13979	0	5	66	436	1318	2784	3652	3554	1790	356	18	0	
	CDH80	5306	0	0	5	76	391	1092	1505	1493	666	77	1	0	
Wind	WSAvg	2.8	3.3	3.6	3.9	3.7	2.8	2.6	2.4	2.1	2.0	2.1	2.4	2.8	
Precipitation	PrecAvg	54.9	4.8	4.5	5.7	4.4	4.9	4.5	5.4	3.6	3.9	3.0	4.4	5.4	
	PrecMax	78.0	9.6	14.6	12.2	14.0	11.9	11.1	19.3	9.1	8.9	6.9	10.8	12.6	
	PrecMin	38.8	0.9	0.8	2.2	0.9	0.8	0.6	1.5	1.3	1.4	0.0	1.8	0.7	
	PrecStd	10.1	2.1	2.4	2.5	2.6	2.6	2.6	3.4	1.8	2.0	1.8	2.0	2.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.4	73.6	79.8	85.8	89.8	94.0	95.3	95.7	93.4	86.4	76.6	69.6	
		MCWB	59.7	61.6	61.7	65.7	71.1	73.0	75.6	74.2	70.6	70.9	63.4	61.0	
	2%	DB	62.9	67.5	75.4	82.3	86.9	91.2	92.4	92.6	90.2	81.7	71.3	64.4	
		MCWB	56.4	58.7	59.8	63.7	69.1	73.1	75.1	74.2	71.1	66.8	59.8	57.9	
	5%	DB	58.2	62.6	71.1	79.4	84.5	88.8	90.2	90.4	87.3	78.0	67.3	59.9	
		MCWB	52.5	53.2	57.3	62.1	68.2	72.1	74.7	73.9	71.1	64.9	58.0	55.0	
	10%	DB	54.2	57.5	66.6	75.8	81.7	86.4	88.0	88.0	83.9	74.2	63.7	56.1	
		MCWB	49.3	50.2	55.0	60.8	67.3	71.2	74.0	73.3	69.8	63.5	55.9	51.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.9	65.1	64.5	69.1	74.2	77.3	79.1	79.0	76.3	74.5	67.0	63.8
			MCDB	66.3	69.3	74.3	81.2	82.7	87.3	89.0	88.1	85.1	81.3	71.3	67.3
2%		WB	57.7	60.2	62.2	66.9	72.2	75.8	77.6	77.3	74.5	70.9	63.9	59.2	
		MCDB	61.9	66.7	71.4	77.7	82.1	85.7	87.4	86.5	82.9	76.1	68.3	62.4	
5%		WB	54.5	55.5	60.0	64.9	70.8	74.6	76.5	76.1	73.1	68.0	60.8	56.0	
		MCDB	57.3	59.8	68.1	74.1	80.5	84.1	85.8	85.1	81.0	74.9	65.4	59.2	
10%		WB	50.2	51.6	57.4	63.2	69.2	73.3	75.4	75.1	71.8	65.5	57.3	52.1	
		MCDB	53.2	55.3	64.5	71.5	77.8	82.1	84.2	83.6	79.6	71.5	62.2	55.4	
Mean Daily Temperature Range		5% DB	MDBR	17.0	18.3	20.7	22.5	21.0	19.8	18.2	19.1	20.2	21.4	20.6	16.7
			MCDBR	20.2	24.9	26.2	27.2	23.9	23.1	21.0	22.1	23.8	25.3	25.2	19.9
	MCWBR		14.2	15.5	13.5	11.7	9.0	7.7	6.8	6.9	7.7	11.2	14.5	13.6	
	5% WB	MCDBR	15.9	19.5	21.5	21.5	20.1	18.7	18.0	18.3	18.9	19.1	18.1	16.6	
		MCWBR	12.9	14.9	12.9	10.8	8.4	7.5	6.7	6.9	7.2	9.8	12.7	13.3	
Clear-Sky Solar Irradiance	taub	0.311	0.319	0.351	0.401	0.453	0.487	0.519	0.528	0.422	0.366	0.330	0.312		
	taud	2.510	2.479	2.423	2.277	2.157	2.099	2.039	1.991	2.274	2.427	2.490	2.527		
	Ebn at Noon	282	293	291	280	266	256	246	241	264	272	272	274		
	Edh at Noon	26	30	34	42	48	51	54	55	40	31	26	24		
All-Sky Solar Radiation	RadAvg	710	891	1260	1628	1806	1936	1825	1719	1464	1162	841	616		
	RadStd	62	114	116	111	136	123	110	87	137	146	93	62		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+248	+151

Nomenclature: See separate page